Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher

```
% Facts
sentence(the_cat_sits).
sentence(the_dog_runs).

% Query
% ?- sentence(X).
```

Output:

```
X = the_cat_sits ; X = the_dog_runs ; false.
```

Exercise 2: Compound Term Unification

```
% Facts
likes(john, food(pizza, spicy)).
likes(mary, food(pizza, sweet)).

% Query
% ?- likes(john, food(pizza, T)).
```

Output:

```
T = spicy.
```

Exercise 3: Conflict Resolution

```
% Facts
rule1(climate_change).
rule2(climate_change).

% Rule
conflict(X) :- rule1(X), rule2(X).

% Query
% ?- conflict(climate_change).
```

Output:

```
true.
```